12 The points A, B, C and D lie in order on a straight line.

$$AB : BD = 1 : 5$$
$$AC : CD = 7 : 11$$

Work out $AB : BC : CD$

Make the whole length AD equal in both ratios.

1 + 5 = 6 parts; 7 + 11 = 18 parts.

Multiply 1 : 5 by 3 to give 3 : 15.

1 mark
Use the same whole length of 18 parts

AB	BD
3	15

AC	CD
7	11

AB	BC	CD
3	4	11

AB = 3

AC = 7

BC = 7 − 3 = 4

CD = 11

1 mark
Subtract AB from AC

1 mark
AB : BC : CD = 3 : 4 : 11

3 : 4 : 11

(Total for Question 12 is 3 marks)